MD. SAMIUL ISLAM MOBIN



Electrical & Electronic Engineer | VLSI & Semiconductor Enthusiast | Researcher

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Mirpur-2, Dhaka, Bangladesh

PROFESSIONAL SUMMARY

Motivated Electrical and Electronic Engineer with high academic excellence and a strong foundation in VLSI and semiconductor design, analog and digital circuit design, embedded systems and AI prompt engineering. Experienced in hands-on industrial training across IC design and power plant engineering. Demonstrated ability to apply theoretical knowledge to real-world engineering projects involving IoT, power electronics, and robotic systems. Seeking opportunities to contribute to innovative engineering teams in VLSI, embedded systems, or automation.

EDUCATION

Bachelor of Electrical & Electronic Engineering *2022 – 2026*

Daffodil International University, Dhaka | Faculty of Engineering

CGPA: 3.96 / 4.0

Higher Secondary Certificate – Science *Passing Year: 2020*

Rajshahi Govt. City College | GPA: 5.0 / 5.0

Secondary School Certificate – Science *Passing Year: 2018*

Daxmin Mainam High School & College | GPA: 5.0 / 5.0

EXPERIENCE & TRAINING

Industrial Training – VLSI & IC Design *2025*

ULKASEMI Pvt. Ltd., Tejgaon-Gulshan Link Road, Dhaka

- Completed structured training in Analog Design, Digital Design, and Design Verification workflows.
- Gained practical experience in Custom Layout techniques and Digital Circuit Design using industry-standard Cadence EDA tools.

Industrial Training – Power Plant Engineering *2025*

Ghorarshal Power Station, Palash, Narsingdi

- Trained in power generation processes including operation and maintenance of power plant equipment.
- Performed electrical system analysis, load management, HMI, SCADA and followed plant safety procedures.

Training – Sales & Marketing *2024*

Skill Jobs (Online)

- Acquired knowledge of sales strategies, customer relationship management, digital marketing, and market research techniques.

General Member – DIU Robotics Club & DIU EEE Club *2022*

Daffodil International University

- Participated and contributed in robotics workshops covering fundamentals, sensor and actuator integration, and microcontroller programming, circuit analysis, teamwork, event coordination, and leadership.

RESEARCH & PUBLICATIONS

[IEEE] A Low-Cost, Ultra Low-Power, and Scalable LoRa-Based Weather Balloon for Atmospheric Monitoring: A Bangladesh Perspective *December 2025*

11th International Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE 2025)

Authors: Md. Tanvir Hassan, Md. Samiul Islam Mobin, Sourav Karmokar, Dr. Tama Fouzder

Doi: *doi.org/10.1109/WIECON-ECE69386.2025.11526116*

[IEEE] A Software-Defined Digital Twin Framework for Enhanced Tele-Laboratory Manipulation in Bio-Hazardous and High-Risk Environments *April 2026*

2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN 2026)

Authors: Md. Tanvir Hassan, Sourav Karmokar, Md. Samiul Islam Mobin, Kazi Mubashir Tazwar, Jafry Anam Sirat, Rahul Islam Ove

Doi: *doi.org/10.1109/QPAIN69676.2026.11546531*

[Conference] A Novel Hexagonal Shaped Suspended-Core Microstructured Fiber for Mid-Infrared Supercontinuum Generation
August 2026

2026 IEEE International Conference on Power, Electrical, Electronics and Industrial Applications (PEEEIACON 2026)

Authors: Shourov Paul, Md. Samiul Islam Mobin, Abu Sayem, Rakayet Rafi, M R Karim

Status: *Presented; IEEE Xplore publication forthcoming*

[Conference] ULTRAFLOAT: An ESP32-S3-Based Non-Contact Water Level Sensing and Pump Control Platform with Dynamic Speed-of-Sound Compensation *September 2026*

2026 IEEE 5th International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON 2026)

Authors: Shourov Paul, Md. Samiul Islam Mobin, Abu Sayem, M R Karim

Status: *Presented; IEEE Xplore publication forthcoming*

PROJECTS

Software-Defined Digital Twin Framework for Tele-Laboratory Manipulation in Bio-Hazardous and High-Risk Environments

- Built a 4-DOF servo-driven robotic arm with a custom gripper and Arduino-based control as a research platform for pick-and-place operations and inverse kinematics.

Low-Cost LoRa-Based Radiosonde for Atmospheric Sounding

- Developed a power-efficient weather balloon payload using ESP32, BME280, and LoRa with dual data logging (onboard SD card + ground station) for real-time atmospheric data acquisition.

12V, 2A Isolated SMPS Design

- Engineered a flyback converter-based SMPS using the DK124 controller IC, achieving 87% efficiency with stable output regulation.

Smart IoT & Solar-Powered Irrigation System

- Designed an off-grid automated irrigation system using ESP32, Blynk IoT, and solar energy harvesting, with remote monitoring and control of irrigation schedules.

ULTRAFLOAT – IoT-Based Water Level Monitoring & Automated Pump Control System | Final Year Capstone Project (EEE400)

- Designed a two-unit ESP32-S3 IoT system with JSN-SR04M ultrasonic and AHT20 sensors, a custom SMPS, a WebSocket live dashboard, and a temperature-compensated distance algorithm ($V = 331.3 + 0.606 \times T$ m/s) for accurate water level monitoring and automated pump control.

TECHNICAL SKILLS

Circuit & IC Design: Analog & Digital Circuit Design, VLSI Design (Cadence), Custom Layout, SMPS / Flyback Design, PCB Schematic
EDA & Engineering Tools: MATLAB / Simulink, AutoCAD Electrical, Proteus Design Suite, Arduino IDE, PLC Programming
Programming Languages: C, C++, Python, HTML
Embedded & IoT Systems: ESP32, Arduino, LoRa, Blynk IoT, Sensor Integration, Microcontroller Programming
Domain Knowledge: Renewable Energy Systems, Wireless Communication, Robotics & Automation, Power Plant Engineering, Data Analysis & Troubleshooting
Productivity Software: MS Office (Word, Excel, PowerPoint), Canva, Video Editing
Soft Skills: Leadership, Technical Communication, Teamwork, Event Coordination, Problem Solving
Languages: Bangla (Native), English (Fluent), Hindi (Intermediate)

ACHIEVEMENTS & CERTIFICATIONS

- **Outstanding Performance – Patent Idea Competition 2025** — *Faculty of Graduate Studies, DIU, 2025*
- **Certificate of Achievement – VLSI Competition** — *MIST Robotics Club, Military Institute of Science & Technology, 2025*
- **DIU-DoR (Research Project) Poster Presentation** — *Division of Research, Daffodil International University, 2025*
- **AI+ Foundation™ – AI CERTs Certification** (Verify) — *AI CERTs, Oct 2026 (valid till Oct 2027)*
- **AI Prompt Engineering at Agent X – Microsoft Netcom Certification** — *Netcom Learning Bangladesh, 2025*
- **DIU Growing Star Program** — *Department of GED, Daffodil International University, 2022*

REFERENCES

Dr. Mohammad Rezaul Karim
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